Math · Theory

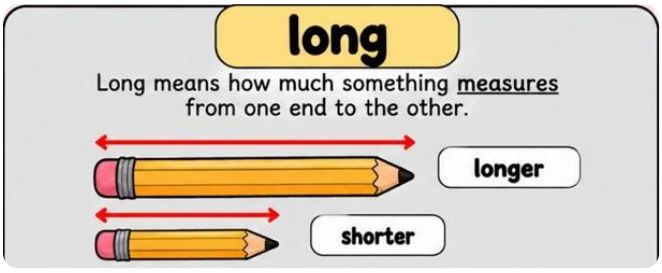
Year 2 · 3rd term · Length, perimeter, 3D solids, data, probability, multiplication and division

1

UNIT 8 · LESSON 1 OF 7

Words to compare and measure

What is length?



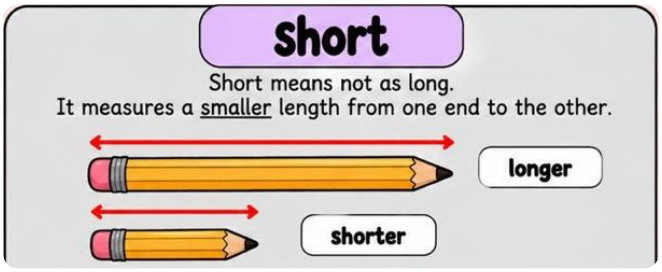
Two pencils

Length is how long something is, from one end to the other.

Long means how much something measures from one end to the other: “The top pencil is long.”

Units — we measure length in centimetres (cm), metres (m) and kilometres (km).

Short, longer and shorter



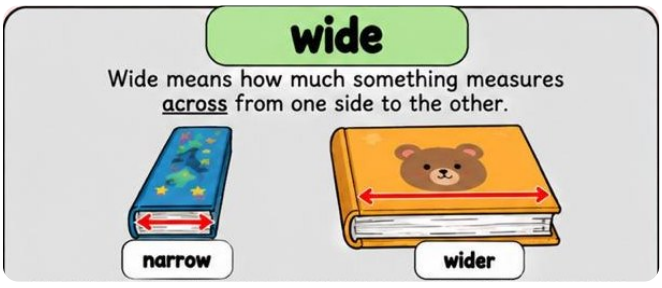
Longer and shorter

Short means not as long: it measures a smaller length from one end to the other. “The bottom pencil is short.”

Longer — the top pencil is longer.

Shorter — the bottom pencil is shorter.

Wide and narrow



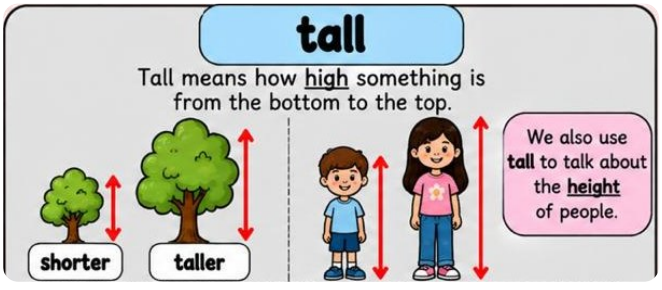
A narrow book and a wide book

Wide means how much something measures across: “The big book is wide.”

Narrow means not wide: the thin book is narrow.

Wider — the big book is wider.

Tall and height



Trees and children

Tall means how high something is, from the bottom to the top: “The big tree is tall.”

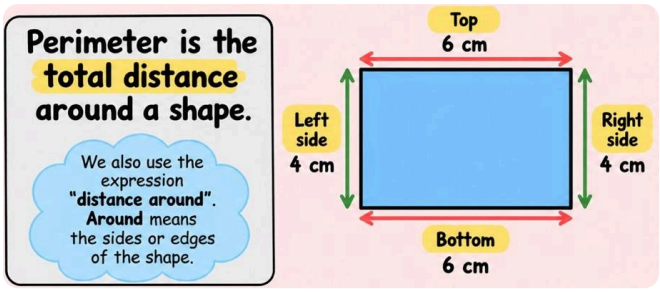
Taller and shorter — the big tree is taller; the small tree is shorter.

Height is how tall someone or something is. We also use tall to talk about the height of people: “The girl is tall.”

UNIT 8 · LESSON 2 OF 7

2 Perimeter

What is perimeter?



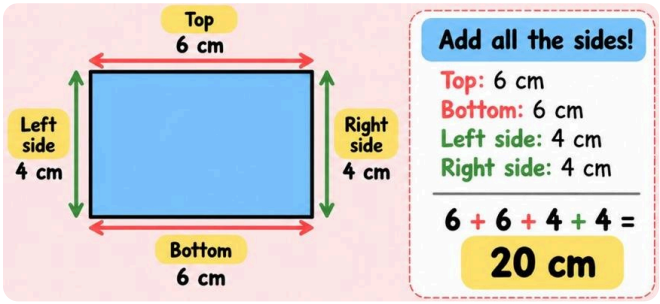
Perimeter: the distance around

Perimeter is the total distance around a shape.

A side is one of the straight lines of a flat shape. A rectangle has 4 sides.

Around means along all the sides of the shape. We also say "distance around".

Add all the sides!



Top, bottom, left side, right side

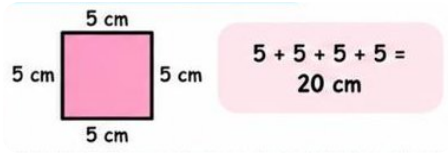
Top and bottom — 6 cm + 6 cm.

Left side and right side — 4 cm + 4 cm.

Perimeter — 6 + 6 + 4 + 4 = 20 cm.

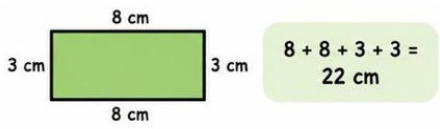
The unit — always write the number and the unit: 20 cm.

More shapes



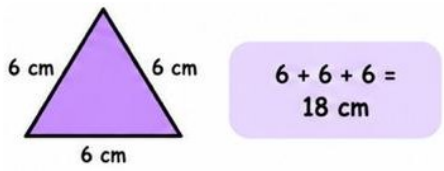
square

Square — 5 + 5 + 5 + 5 = 20 cm.



rectangle

Rectangle — 8 + 8 + 3 + 3 = 22 cm.



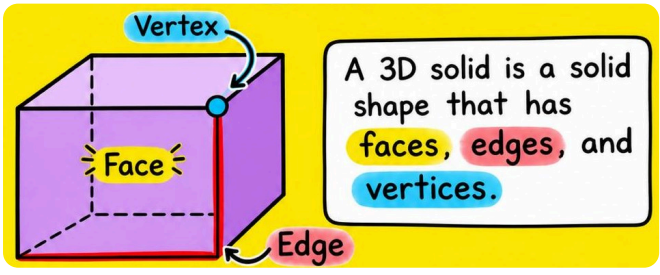
triangle

Triangle — 6 + 6 + 6 = 18 cm.

UNIT 6 · LESSON 3 OF 7

3 3D solids

What is a 3D solid?



A cube

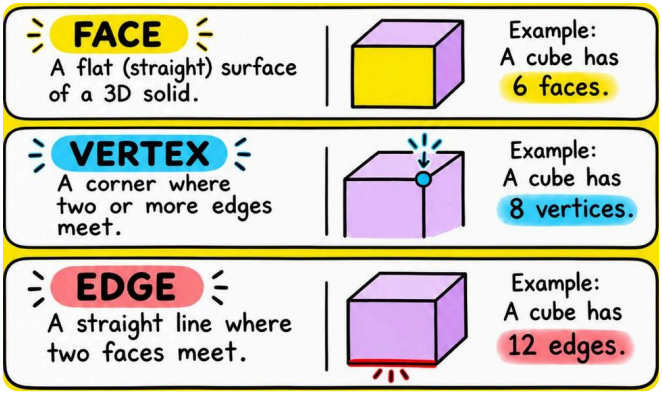
A 3D solid is a solid shape that has faces, edges and vertices. A cube is a 3D solid.

A face is a flat (straight) surface of a 3D solid.

An edge is a straight line where two faces meet.

A vertex is a corner where two or more edges meet. More than one vertex: vertices.

The cube



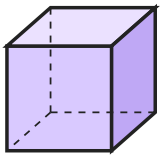
A cube has 6 faces, 8 vertices and 12 edges

Faces — a cube has 6 faces.

Vertices — a cube has 8 vertices.

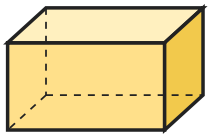
Edges — a cube has 12 edges.

Cube, cuboid and pyramid



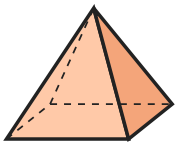
cube

Cube — 6 faces, 12 edges, 8 vertices.



cuboid

Cuboid — 6 faces, 12 edges, 8 vertices. It looks like a cube, but its edges are not all the same length.

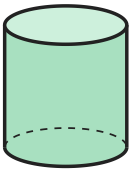


pyramid

Pyramid — 5 faces, 8 edges, 5 vertices.

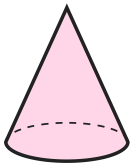
Flat or curved?

A curved surface is round, not flat. A cylinder, a cone and a sphere have a curved surface.



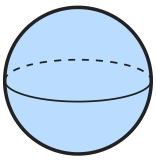
cylinder

Cylinder — 2 flat faces and 1 curved surface.



cone

Cone — 1 flat face and 1 curved surface.

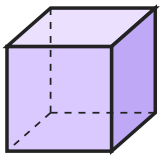


sphere

Sphere — 1 curved surface. No flat face, no edges, no vertices.

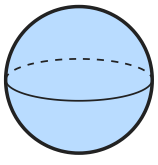
Stay still or roll?

Roll means to move by turning over and over, like a ball.



cube

Flat faces — the cube can stay still on top of other flat surfaces. It cannot roll.

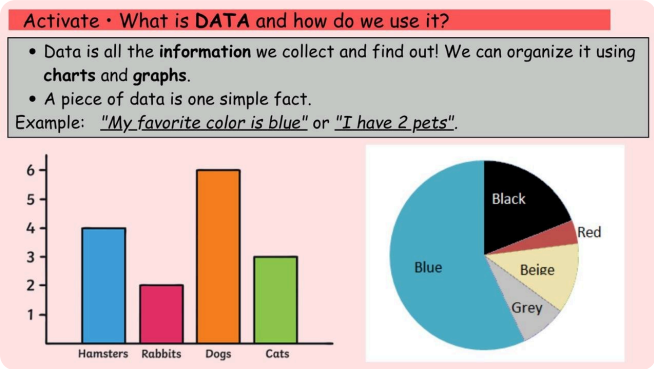


sphere

Curved surface — the sphere can roll away.

Data and graphs

What is data?



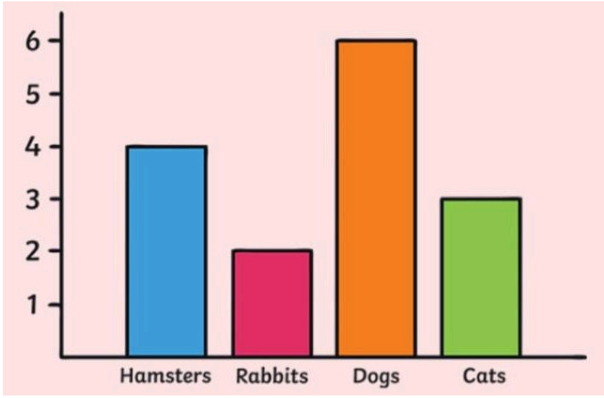
Data in two charts

Data is all the information we collect and find out. We organize it using charts and graphs.

A piece of data is one simple fact: “My favorite color is blue.” “I have 2 pets.”

Charts and graphs — the two pictures organize data.

A bar graph



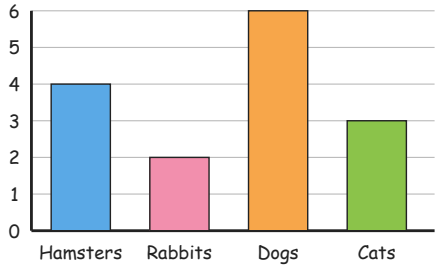
A bar graph about pets

A bar graph shows data with bars. The taller the bar, the more there are.

Each bar — one group: hamsters, rabbits, dogs or cats.

How many — look at the top of the bar and read the number on the side.

Read the bar graph



The bar graph of the pets

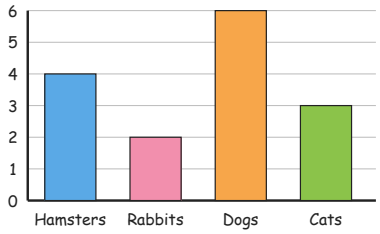
How many — hamsters 4, rabbits 2, dogs 6, cats 3.

The most — dogs: the tallest bar.

The fewest — rabbits: the shortest bar.

Make a bar graph

Hamsters	Rabbits	Dogs	Cats
4	2	6	3



From the table to the graph

The table — shows the number of each group.

Color each bar — up to its number.

Dogs: 6 — color the dogs bar up to 6.

5

UNIT 5 · LESSON 5 OF 7

Probability

What is probability?

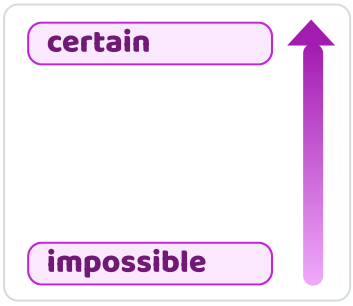


From top to bottom

Probability is thinking about if something will happen, won't happen, or will maybe happen.

The arrow — at the top, it will happen; at the bottom, it won't happen.

Certain and impossible

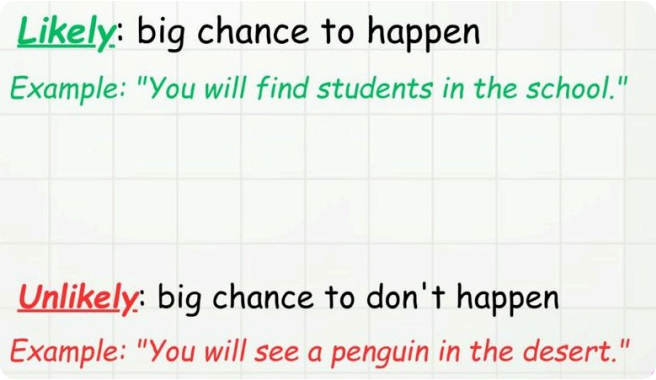


The top and the bottom

Certain means we know it will happen: "I will eat dinner tonight."

Impossible means we know it won't happen: "I will eat rocks for dinner." "A unicorn carries you to bed."

Likely and unlikely

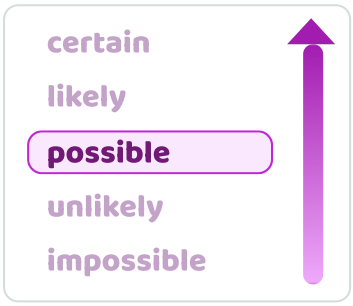


Likely and unlikely

Likely means big chance to happen: “You will find students in the school.”

Unlikely means small chance to happen: “You will see a penguin in the desert.”

Possible



All five, from the top

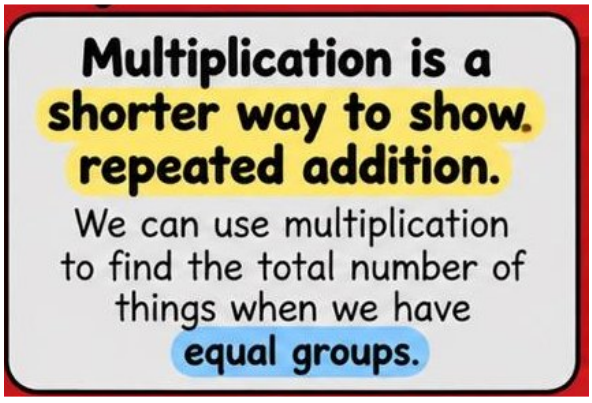
Possible means it will maybe happen: “I will eat spaghetti for dinner.” “I will read a book.”

From the top — certain, likely, possible, unlikely, impossible.

UNIT 10 · LESSON 6 OF 7

6 Multiplication: making groups

What is multiplication?



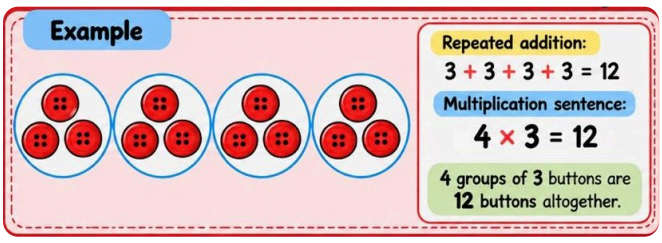
Multiplication

Multiplication is a shorter way to show repeated addition. We use it to find the total when we have equal groups.

Equal groups have the same amount in each group.

Repeated addition is adding the same number again and again: 3 + 3 + 3 + 3.

4 groups of 3



4 groups of 3 buttons

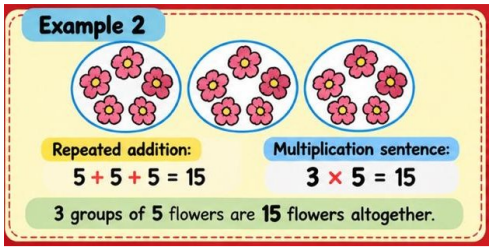
Repeated addition — 3 + 3 + 3 + 3 = 12.

A multiplication sentence is the math written with numbers and signs: 4 × 3 = 12. 3 + 3 + 3 + 3 = 12 is an addition sentence.

4 × 3 — 4 groups, 3 buttons in each group.

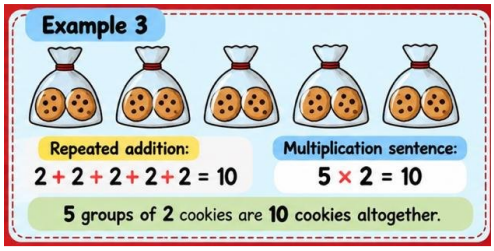
Altogether — 4 groups of 3 buttons are 12 buttons altogether.

More groups



3 groups of 5 flowers

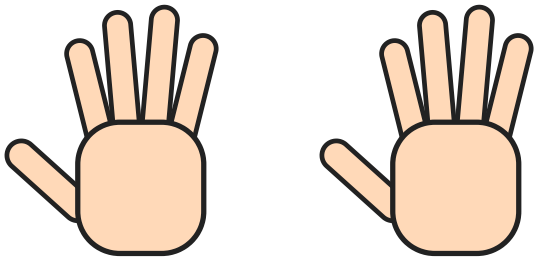
Flowers — 5 + 5 + 5 = 15, so 3 × 5 = 15. 3 groups of 5 flowers are 15 flowers altogether.



5 groups of 2 cookies

Cookies — 2 + 2 + 2 + 2 + 2 = 10, so 5 × 2 = 10. 5 groups of 2 cookies are 10 cookies altogether.

Our hands



2 hands

2 hands — 5 fingers on each hand.

Repeated addition — 5 + 5 = 10.

Multiplication sentence — 2 × 5 = 10.

7

UNIT 10 · LESSON 7 OF 7

Division: equal sharing

What is division?



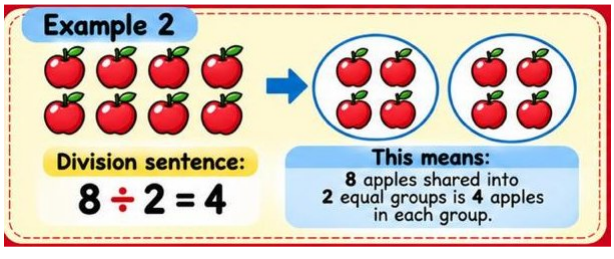
Mike and Laura share 10 oranges

Division is a way to make equal groups of a number. We share a total into groups that have the same amount.

Share equally means everybody gets the same amount.

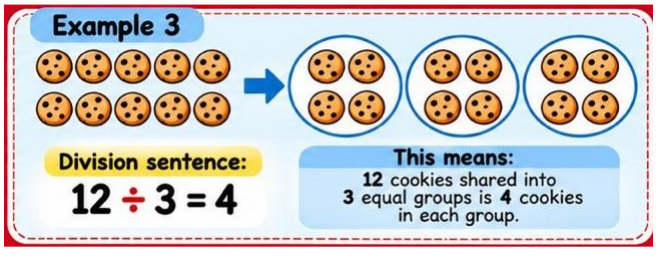
Division sentence — 10 ÷ 2 = 5: 10 oranges shared into 2 equal groups is 5 oranges in each group.

More sharing



8 apples, 2 groups

Apples — 8 ÷ 2 = 4: 8 apples shared into 2 equal groups is 4 apples in each group.



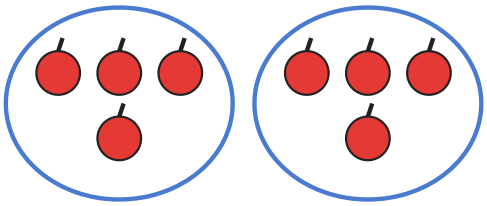
12 cookies, 3 groups

Cookies — 12 ÷ 3 = 4: 12 cookies shared into 3 equal groups is 4 cookies in each group.

Even and odd: into 2 groups

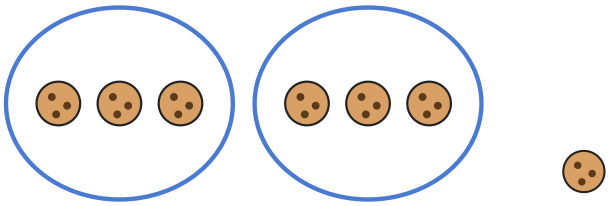
Even numbers (0, 2, 4, 6, 8, 10, 12...) can be shared equally into 2 groups.

Odd numbers (1, 3, 5, 7, 9...): when we share them between 2, 1 is left.



8 apples, 2 groups

8 ÷ 2 = 4 — 4 in each group. Nothing is left.



7 cookies, 2 friends

7 between 2 — 3 each, and 1 is left.